

Kepler's equation and its application to the patched conic approximation

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Abstract

Kepler's equation provides the fundamental link between time and orbital position for bodies moving under a central inverse-square gravitational field [1, 2, 3]. Owing to its transcendental nature, the equation admits no closed-form solution and must be solved numerically. This project studies the application of Kepler's equation to orbit propagation within the patched conic approximation [4, 5]. Using Two-Line Element (TLE) data for the UM5-EOSAT satellite, analytical Keplerian propagation is compared against numerical two-body integration. Discrepancies between the two approaches are quantified, and the sensitivity of the results to numerical accuracy and precision is examined. The methodology is further extended to translunar Hohmann transfers and restricted three-body dynamics in the Earth–Moon system.

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1 Introduction

Kepler’s equation forms the backbone of classical astrodynamics, providing the direct link between time and orbital position for motion governed by a central inverse-square gravitational field [1, 2]. Despite the apparent simplicity of the two-body problem, the equation is transcendental and requires numerical solution in practical applications.

In mission analysis, Keplerian propagation is often used as a fast and reliable baseline, while numerical integration is treated as a higher-fidelity reference. Understanding the level of agreement between these approaches is therefore essential, particularly when Keplerian models are extended to multi-body environments through the patched conic approximation [4].

This project focuses on quantitative validation rather than theoretical derivation. Analytical propagation based on Kepler’s equation is directly compared against numerical two-body integration using consistently reconstructed initial conditions.. The methodology is first applied to a real low Earth orbit satellite (UM5-EOSAT), then extended to Earth–Moon Hohmann transfers, and finally examined under restricted three-body dynamics to identify the limits of Keplerian and patched conic modeling.

2 Keplerian Motion and Kepler’s Equation

Keplerian motion arises as the exact solution of the Newtonian two-body problem [3, 2]. The orbital geometry is fully described by a set of orbital elements, while the time evolution is governed by Kepler’s equation [1]. The transcendental equation is solved numerically using iterative root-finding methods, as no closed-form solution exists

2.1 Two-Body Problem

Assuming a satellite of negligible mass orbiting a central body of mass M , the equation of motion is

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r}, \quad (1)$$

where $\mu = GM$ is the gravitational parameter.

2.2 Orbital Elements

An elliptic orbit is fully characterized by six classical orbital elements: the semi-major axis a , eccentricity e , inclination i , right ascension of the ascending node Ω , argument of periapsis ω , and mean anomaly M at epoch. These elements are commonly provided in satellite catalogues such as TLE files.

2.3 Kepler's Equation

Kepler's equation relates the mean anomaly M to the eccentric anomaly E .

$$M = E - e \sin E, \quad (2)$$

Once E is obtained, the radial distance and true anomaly follow from

$$r = a(1 - e \cos E), \quad (3)$$

$$\tan \frac{\theta}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{E}{2}. \quad (4)$$

3 Patched Conic Approximation

The patched conic approximation models complex trajectories as a sequence of two-body motions, each valid within a region dominated by a single gravitational body [4, 6]. Boundaries between these regions are commonly defined using the concept of a sphere of influence. Despite neglecting higher-order perturbations, the method remains effective for preliminary mission analysis.

4 UM5-EOSAT Mission Background and Motivation

UM5-EOSAT is a low Earth orbit Earth observation satellite developed by Mohammed V University (UM5) in Morocco. The satellite represents Morocco's first university-developed Earth observation mission and is intended to support remote sensing, academic research, and technology demonstration activities.

UM5-EOSAT was deliberately selected for this study not only due to the availability of orbital data, but also because it is a satellite developed by a university in the author's home country. This choice provides a meaningful real-world context for applying classical astrodynamics tools while highlighting their relevance to national and academic space missions.

5 Application to UM5-EOSAT

5.1 TLE Data and Orbital Reconstruction

Orbital parameters for the UM5-EOSAT satellite were obtained from publicly available NORAD Two-Line Element (TLE) data. TLEs provide mean orbital elements rather than instantaneous osculating elements, requiring reconstruction of the osculating state at the epoch before Keplerian propagation [7, 8]. The TLE-provided mean elements were first converted into an osculating Cartesian state at the epoch. This reconstructed state vector was then used as the common initial condition for both analytical Keplerian propagation and numerical integration.

Table 1: General Characteristics of the UM5-EOSAT Satellite

Parameter	Description
Satellite Name	UM5-EOSAT
Launch Date	August 16, 2024
Launch Vehicle	Falcon 9 (Transporter-11)
Launch Site	Air Force Western Test Range (AFWTR)
Primary Mission	Earth Observation / Remote Sensing
Orbit Type	Low Earth Orbit (LEO)
Operator	Mohammed V University (UM5)

The correct reconstruction of osculating parameters at the epoch is therefore essential before applying Keplerian propagation. Failure to do so can result in significant geometric and temporal inconsistencies.

Parameter	Value
Epoch	Sat 20 Dec 2025 03:59:47 GMT+3
Semi-major axis (m)	6.93226×10^6
Eccentricity	0.0007172
Inclination (deg)	97.7043°
RAAN (deg)	69.8943°
Argument of perigee (deg)	357.861°
Mean anomaly (deg)	2.2577

Figure 1: Orbital Parameters of UM5-EOSAT at Epoch (Derived from TLE)

5.2 Analytical Keplerian Propagation

Starting from the reconstructed epoch state, the satellite position is propagated analytically to a selected future time by solving Kepler’s equation numerically for the eccentric anomaly. Figure 2 shows the UM5-EOSAT orbit in the orbital plane obtained using analytical Keplerian propagation for a propagation time $t_{\text{fin}} = 2010.34$ s. The trajectory is Earth-centered and confined to a single plane, as expected for two-body motion.

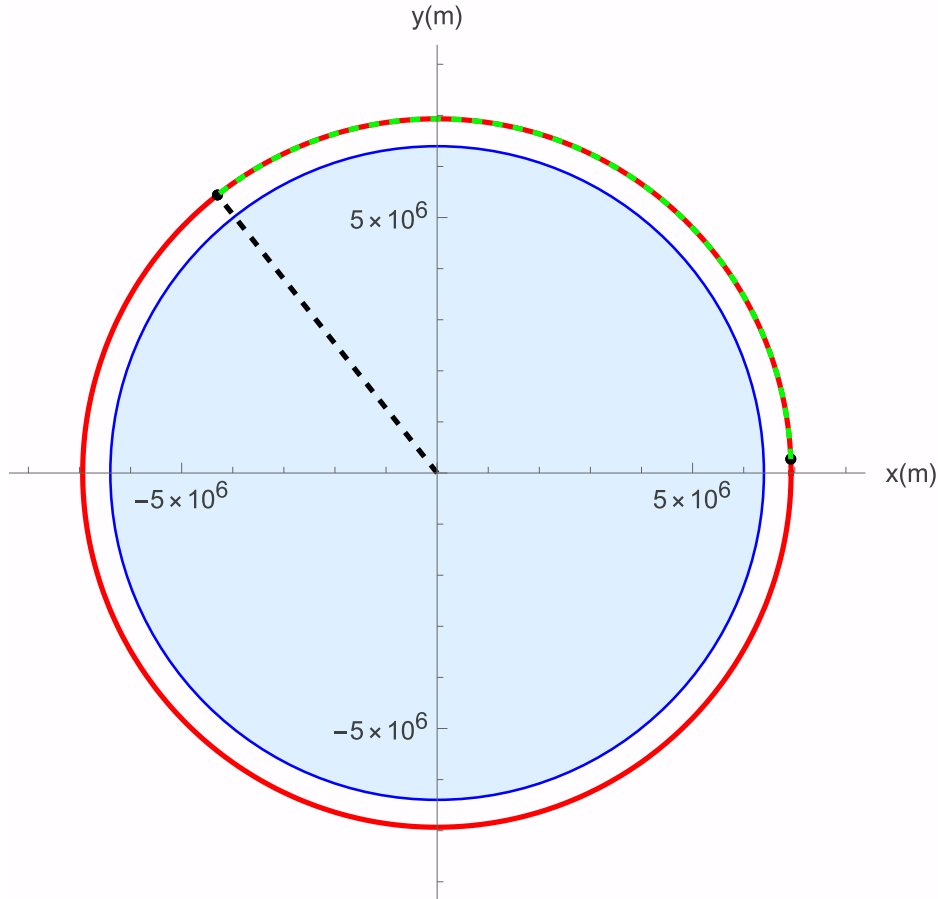


Figure 2: Orbit plane and spacecraft trajectory of UM5-EOSAT obtained using analytical Keplerian propagation for a programatically chosen propagation time $t_{\text{fin}} = 2010.34$ s.

The resulting trajectory is Earth-centered and planar, as expected for ideal two-body motion. The low eccentricity inferred from the TLE data produces a nearly circular orbit, providing a clean test case in which analytical and numerical solutions can be compared without strong sensitivity to initial condition errors. This analytical solution is therefore used as the Keplerian reference model in the subsequent error analysis.

5.3 Numerical Two-Body Integration

The equations of motion were integrated in polar coordinates using `NDSolve`, with the same initial radius and tangential velocity as the analytical solution. This solution serves as a high-fidelity reference for evaluating the analytical approach.

6 Error Analysis

6.1 Overview of Analytical–Numerical Error Behavior

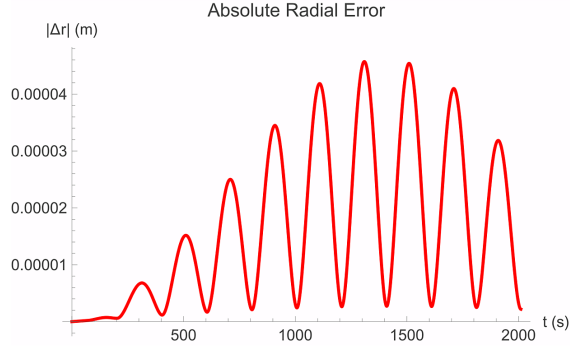
Numerical integration of the two-body equations of motion provides a reference solution against which analytical Keplerian propagation is assessed [9]. Absolute and relative discrepancies in radial distance are defined as

$$\Delta r = |r_{\text{num}} - r_{\text{anal}}|, \quad (5)$$

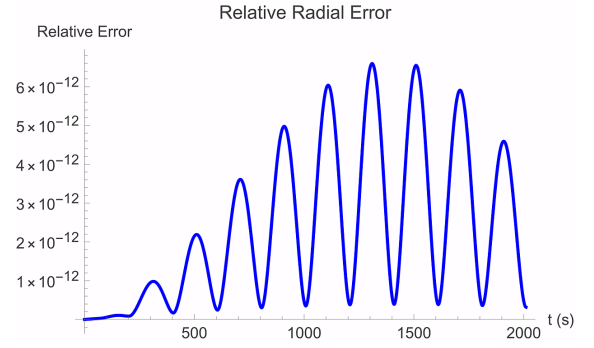
$$\delta r = \frac{|r_{\text{num}} - r_{\text{anal}}|}{r_{\text{anal}}}, \quad (6)$$

with analogous definitions for angular position, where r_{num} is obtained from numerical integration using `NDSolve`, and r_{anal} is obtained from the analytical Keplerian solution.

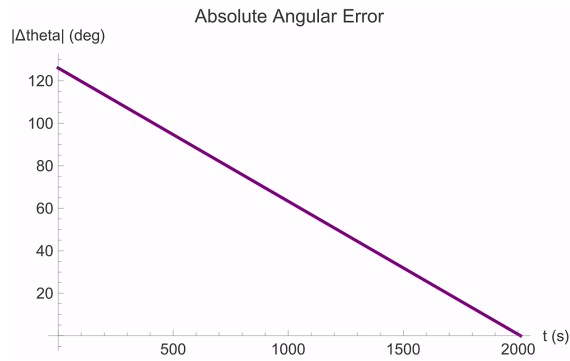
Figure 3 demonstrates excellent agreement between analytical Keplerian propagation and numerical two-body integration over the full propagation interval. Absolute radial errors remain below 5×10^{-4} m, while relative radial discrepancies are bounded at the 10^{-12} level.



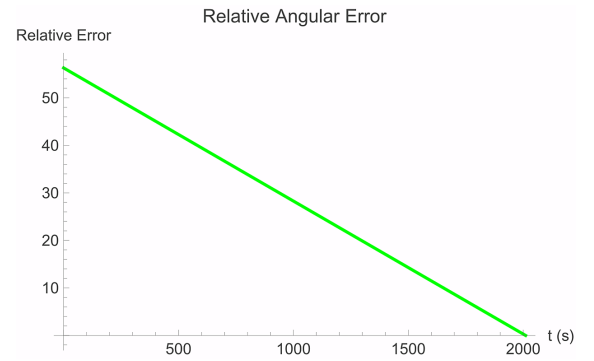
(a) Absolute radial discrepancy



(b) Relative radial discrepancy



(c) Absolute angular discrepancy



(d) Relative angular discrepancy

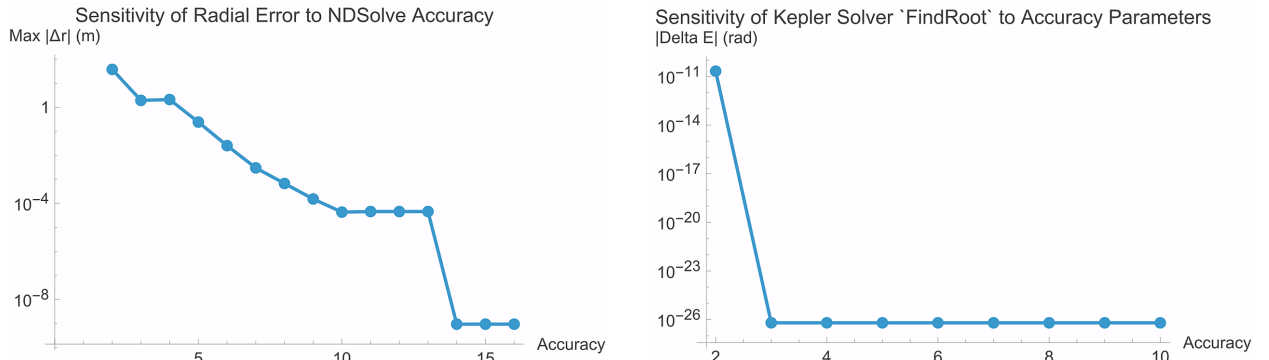
Figure 3: Absolute and relative discrepancies in radial distance and angular position between analytical Keplerian propagation and numerical two-body integration.

Radial errors exhibit a small, bounded oscillation correlated with orbital phase, indicat-

ing numerical truncation effects rather than secular divergence. Angular discrepancies decrease overall and remain bounded without secular growth, reflecting progressive phase alignment once identical initial conditions are enforced. The absence of long-term error growth confirms that analytical Keplerian propagation achieves near machine-precision agreement for low-eccentricity Earth orbits.

6.2 Sensitivity to Numerical Accuracy and Precision

Sensitivity studies reveal that residual discrepancies are governed primarily by numerical integration accuracy rather than by the solution of Kepler’s equation itself. Increasing the `NDSolve` accuracy parameter reduces the maximum radial error by several orders of magnitude until a plateau near 10^{-8} m is reached, indicating a transition to floating-point-limited precision.



(a) Sensitivity of radial error to `NDSolve` accuracy parameters

(b) Sensitivity of Kepler solver (`FindRoot`) to accuracy parameters

Figure 4: Dependence of analytical–numerical discrepancies on numerical accuracy and precision parameters swept from $x = 2$ to $x = 16$.

In contrast, the Kepler equation solver (`FindRoot`) converges rapidly, with eccentric anomaly errors reaching machine precision ($\sim 10^{-26}$ rad) at relatively low accuracy settings. Beyond this point, further refinement produces no measurable improvement, confirming that the analytical solver is not a limiting source of error in the present analysis.

7 Application to Translunar Hohmann Transfer

The patched conic approximation is applied to the classical Earth–Moon Hohmann transfer [1, 4]. The transfer ellipse has apogee at the lunar orbital radius and perigee at the departure orbit. The apogee of the transfer ellipse is therefore,

$$r_a = r_{\text{Moon}} = 3.844 \times 10^8 \text{ m.}$$

The perigee radius is defined as

$$r_p = R_{\oplus} + H_p,$$

where H_p denotes the departure altitude.

This analysis assumes coplanar Earth–Moon orbits, circular lunar motion, and impulsive velocity changes, consistent with the patched conic approximation.

7.1 Transfer Geometry and Travel Time

For a Hohmann transfer ellipse, the semi-major axis is

$$a = \frac{r_p + r_a}{2}, \quad (7)$$

and the orbital period of the full ellipse is

$$T = 2\pi \sqrt{\frac{a^3}{\mu_{\oplus}}}. \quad (8)$$

Since the spacecraft travels only half of the ellipse before reaching lunar distance, the transfer time is

$$t_{\text{Hoh}} = \frac{T}{2}. \quad (9)$$

7.1.1 Dependence on Initial Altitude

Although increasing the perigee altitude reduces the velocity required for injection, it also increases the semi-major axis of the transfer ellipse. Since the orbital period scales as $a^{3/2}$, the resulting transfer time increases with increasing initial altitude. This explains the counterintuitive result that departures from higher orbits lead to longer Earth–Moon travel times, despite requiring smaller impulsive velocity increments [1, 3].

7.2 Velocity Increment for Translunar Injection

The required tangential velocity at perigee of the transfer orbit is obtained from the specific angular momentum,

$$v_p = \frac{h}{r_p}. \quad (10)$$

Subtracting the circular velocity at the departure orbit yields the required injection impulse,

$$\Delta v = v_p - \sqrt{\frac{\mu_{\oplus}}{r_p}}. \quad (11)$$

Numerical evaluation shows that Δv decreases monotonically as the departure altitude increases, in agreement with classical estimates reported by Bate et al. [1].

7.3 Rocket Equation and Burn Characteristics

A representative initial spacecraft mass was assumed for illustrative purposes. The propellant mass required to achieve translunar injection is obtained from the Tsiolkovsky rocket equation,

$$\Delta M = M_{\text{in}} (1 - e^{-\Delta v / v_{\text{exh}}}), \quad (12)$$

where the exhaust velocity is

$$v_{\text{exh}} = I_s g_0. \quad (13)$$

Using a representative specific impulse of $I_s = 311$ s and a thrust level of 100 kN, the corresponding mass flow rate is

$$\dot{m} = \frac{F}{I_s g_0}. \quad (14)$$

The resulting burn duration,

$$\Delta t_{\text{burn}} = \frac{\Delta M}{\dot{m}}, \quad (15)$$

was found to be on the order of a few hundred seconds, which is several orders of magnitude smaller than the transfer period. This confirms the validity of the impulsive-burn approximation for translunar injection [4, 10].

7.4 Departure from Geostationary Orbit

The same analysis was repeated for a spacecraft departing from a geostationary orbit (GEO). While the required Δv and propellant mass are significantly reduced compared to a low-Earth orbit (LEO) departure, the corresponding Hohmann transfer time increases due to the larger semi-major axis.

This highlights the classical trade-off between propulsion requirements and time of flight in trajectory design. GEO departures may be advantageous for missions prioritizing reduced propellant consumption over transfer duration [5].

7.5 Analytical Position Along the Transfer

The spacecraft position along the transfer orbit is obtained analytically by solving Kepler's equation at discrete times,

$$M(t) = nt,$$

where n is the mean motion of the transfer ellipse. The eccentric anomaly is computed numerically and converted into radial distance and true anomaly.

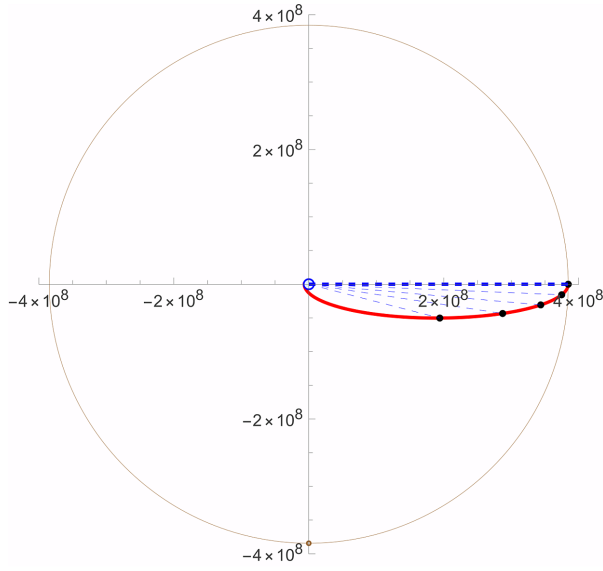
Because the periapsis of the translunar ellipse lies on the negative x -axis, the argument of periapsis is $\omega = \pi$, which must be added to the computed true anomaly to obtain the correct inertial orientation.

Positions evaluated at 24-hour intervals up to arrival at apogee show a smooth, monotonic increase in radial distance, consistent with Keplerian motion.

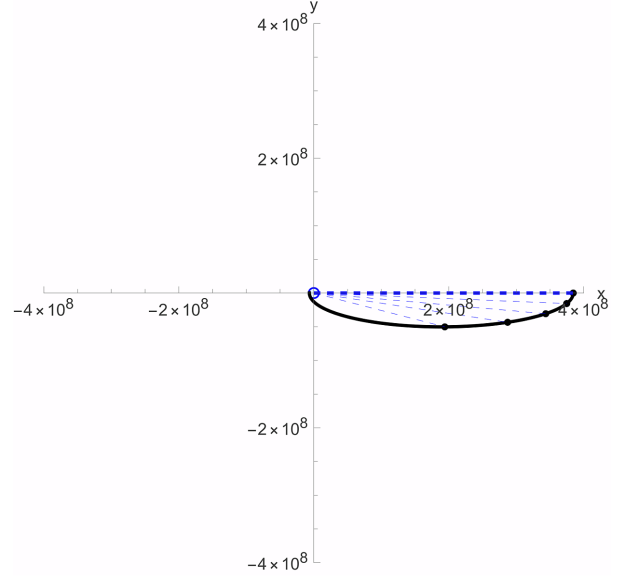
7.6 Numerical Propagation

In parallel, the same trajectory is propagated using numerical integration of the two-body equations of motion in polar coordinates. The integration is initialized at perigee with the analytically computed tangential velocity. Figure 5 compares analytical Keplerian propagation and numerical two-body integration for the Earth–Moon Hohmann transfer. The

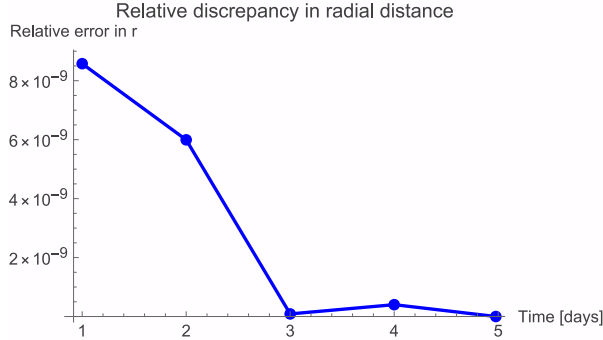
discrete markers along the transfer arc indicate spacecraft positions at successive daily intervals, showing consistent temporal alignment between analytical and numerical trajectories throughout the transfer phase.



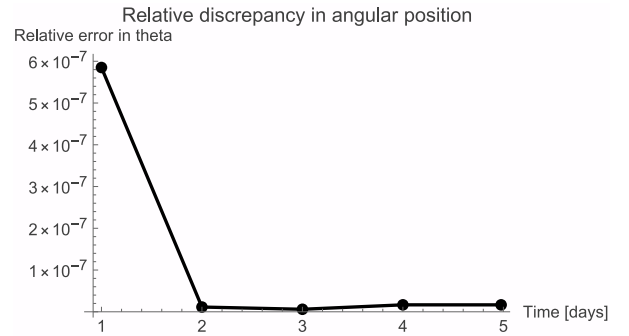
(a) Analytical Hohmann transfer trajectory



(b) Numerical Hohmann transfer trajectory



(c) Relative radial discrepancy



(d) Relative angular discrepancy

Figure 5: Comparison between analytical Keplerian propagation and numerical two-body integration for the Earth–Moon Hohmann transfer.

Figure 5 shows close agreement between analytical Keplerian propagation and numerical two-body integration throughout the Earth–Moon transfer. Discrete markers denote spacecraft positions at successive 24-hour intervals, confirming consistent temporal progression along the transfer arc.

Relative radial discrepancies remain below 10^{-8} , while angular discrepancies rapidly decrease after the initial phase and remain bounded near 10^{-7} . These results validate the patched conic Keplerian model for Earth–Moon transfer analysis up to lunar distance, with

residual differences attributable to numerical precision limits.

7.7 Analytical–Numerical Discrepancy

Relative discrepancies between analytical and numerical solutions are defined as

$$\delta r(t_i) = \frac{|r_{\text{num}}(t_i) - r_{\text{anal}}(t_i)|}{r_{\text{anal}}(t_i)}, \quad (16)$$

$$\delta \theta(t_i) = \frac{|\theta_{\text{num}}(t_i) - \theta_{\text{anal}}(t_i)|}{\theta_{\text{anal}}(t_i)}. \quad (17)$$

These quantities were evaluated at $t_i = 1, 2, 3, 4$ days and at t_{Hoh} . The discrepancies remain extremely small throughout the transfer, with radial errors on the order of 10^{-9} and angular errors on the order of 10^{-7} . This confirms the excellent agreement between analytical Keplerian propagation and numerical two-body integration when consistent initial conditions are employed.

8 Restricted Three-Body Analysis and Collision Window

While the classical Earth–Moon Hohmann transfer provides a useful baseline, it neglects the gravitational influence of the Moon prior to arrival. To investigate the sensitivity of lunar impact conditions to initial velocity, the problem was extended to a restricted three-body formulation in which the spacecraft motion is influenced simultaneously by the Earth and the Moon, while the spacecraft mass remains negligible [11, 12].

Starting from an Earth-centered initial orbit with fixed initial radius $r_0 = R_{\oplus} + H_0$, the spacecraft was injected tangentially with varying initial velocity v_{t0} . For each value of v_{t0} , the equations of motion were numerically integrated under Earth–Moon gravity, and the minimum spacecraft–Moon distance was recorded.

A collision with the Moon was deemed to occur when the minimum distance fell below the lunar mean radius,

$$R_{\text{Moon}} = 1737.4 \text{ km.}$$

Sweeping the initial tangential velocity revealed a narrow and well-defined collision window. Lunar impact trajectories were observed only for velocities in the range

$$10933.3 \text{ m/s} < v_{t0} < 10936.2 \text{ m/s.}$$

Outside this interval, the spacecraft either undershot the Moon, remaining on an Earth-bound orbit, or overshot the lunar sphere of influence and escaped without collision. Figure 6 presents the results of the restricted three-body analysis, highlighting the sensitivity of lunar collision outcomes to the initial tangential velocity. Impact trajectories occur only within a narrow velocity window, $10933.3 \text{ m/s} < v_{t0} < 10936.2 \text{ m/s}$, while slightly lower or higher velocities result in non-collisional flyby trajectories.

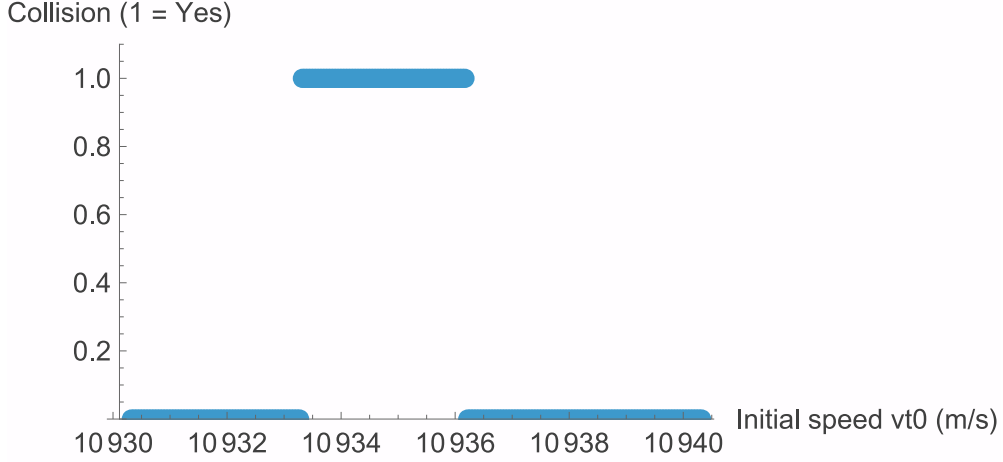


Figure 6: Restricted three-body analysis showing the range of initial tangential velocities leading to lunar collision. Impact trajectories occur only within the narrow velocity window $10933.3 \text{ m/s} < v_{t0} < 10936.2 \text{ m/s}$.

Figure 6 illustrates the extreme sensitivity of lunar impact outcomes to initial tangential velocity under restricted three-body dynamics. Lunar collision occurs only within a narrow velocity interval of approximately 3 m/s, while slightly lower or higher velocities result in flyby or escape trajectories. This sharp transition reflects the nonlinear nature of Earth–Moon dynamics and demonstrates the limitations of Keplerian and patched conic models for precise lunar targeting.

9 Discussion

The results demonstrate that analytical Keplerian propagation provides near machine-precision agreement with numerical two-body integration when consistent initial conditions are employed. For the UM5-EOSAT orbit, both radial and angular discrepancies remain bounded and non-accumulative, confirming the reliability of Kepler’s equation for low-eccentricity Earth orbits.

Sensitivity analysis shows that numerical integration accuracy, rather than the solution of Kepler’s equation, dominates residual discrepancies once solver convergence is achieved. This highlights the importance of integration settings when numerical propagation is used as a reference solution.

For translunar trajectories, analytical Hohmann transfer modeling accurately reproduces numerical two-body results up to lunar distance, validating the patched conic approximation for preliminary Earth–Moon mission design. However, restricted three-body analysis reveals a narrow lunar collision window, showing that small velocity deviations can lead to qualitatively different outcomes. This underscores the necessity of higher-fidelity multi-body modeling for precise lunar targeting.

10 Conclusion

This project confirms that Keplerian propagation based on Kepler’s equation offers an efficient and highly accurate framework for orbit analysis under two-body dynamics. When combined with the patched conic approximation, it provides reliable predictions for Earth–Moon transfer trajectories up to lunar encounter.

Numerical sensitivity studies show that integration accuracy governs residual errors once analytical solver convergence is achieved. Restricted three-body analysis further demonstrates the limitations of Keplerian models near the Moon, where small velocity deviations lead to qualitatively different outcomes. Future work may incorporate perturbations and higher-fidelity dynamical models to extend the analysis to precision lunar mission design.

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